

AI Platform

Overview of Transformer

Mingyu Kim,
Apr, 14th, 2026

Stitching all the pieces together with an example

A cute teddy bear is reading.

Stitching all the pieces together with an example

A

cute

teddy bear

is

reading

.

Stitching all the pieces together with an example

[BOS]	A	cute	teddy bear	is	reading	.	[EOS]
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Stitching all the pieces together with an example



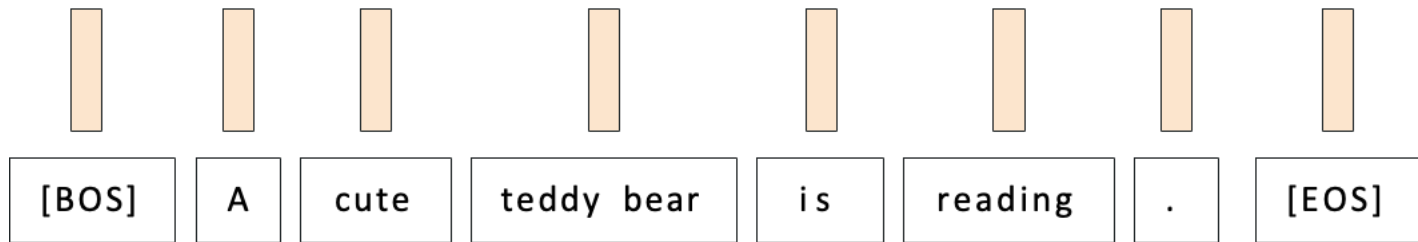
Stitching all the pieces together with an example



Stitching all the pieces together with an example

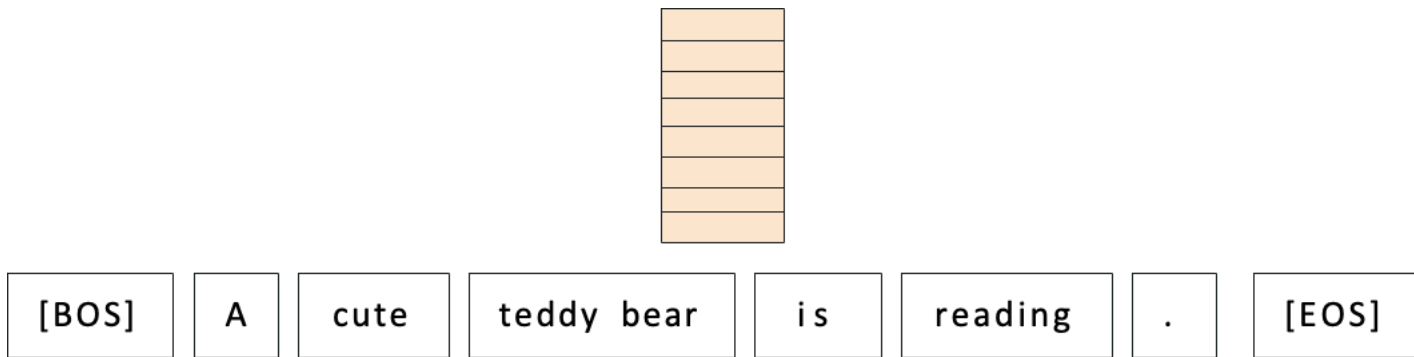


Stitching all the pieces together with an example



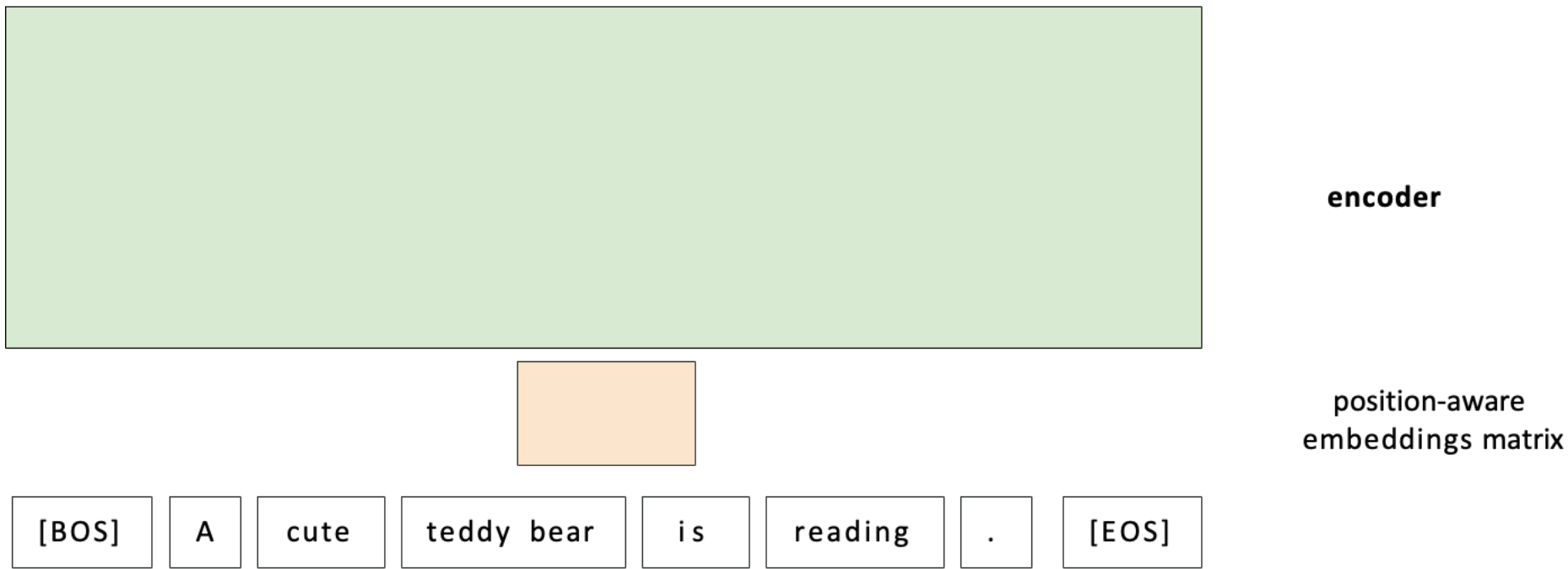
**position-aware
embeddings**

Stitching all the pieces together with an example

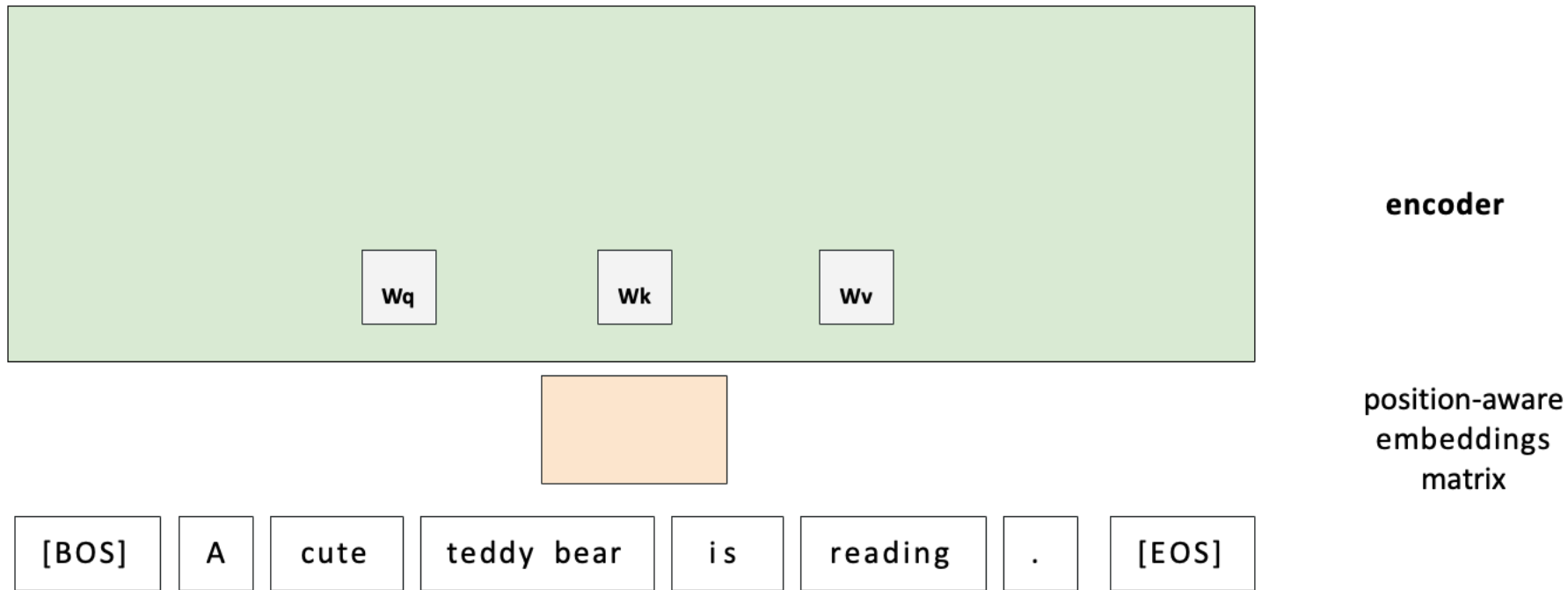


**position-aware
embeddings
matrix**

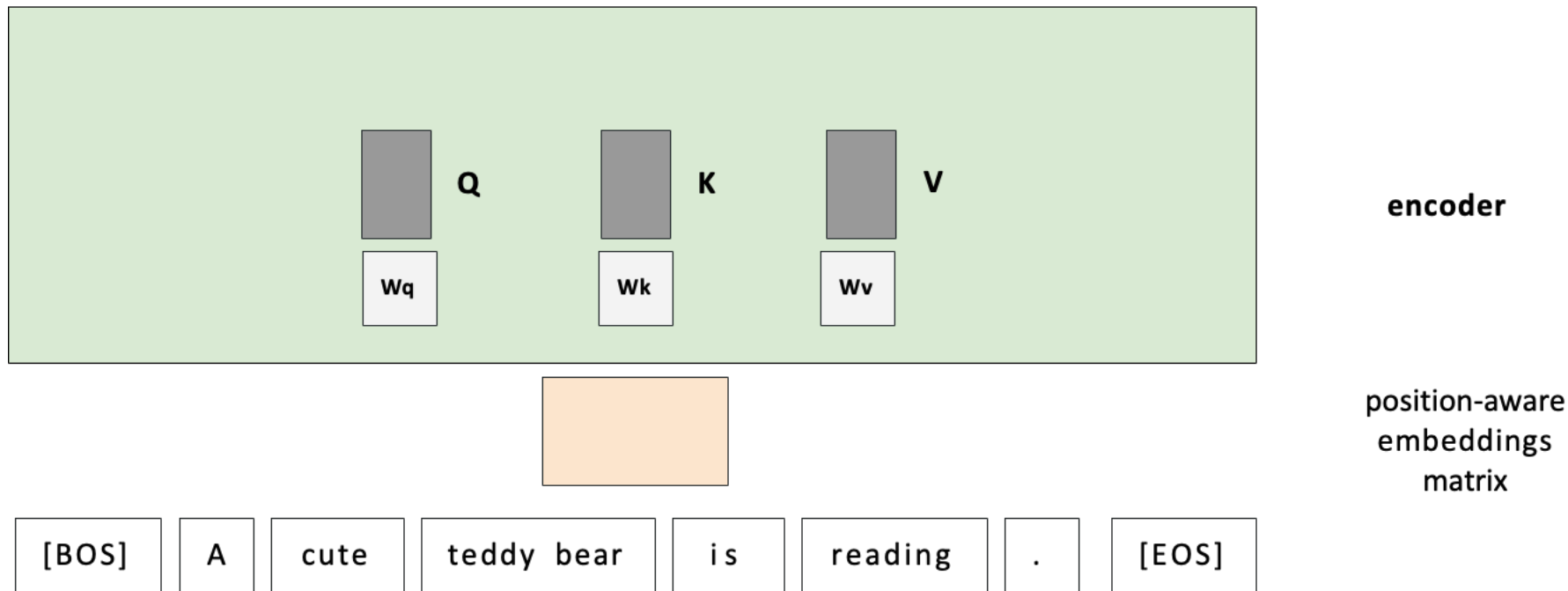
Stitching all the pieces together with an example



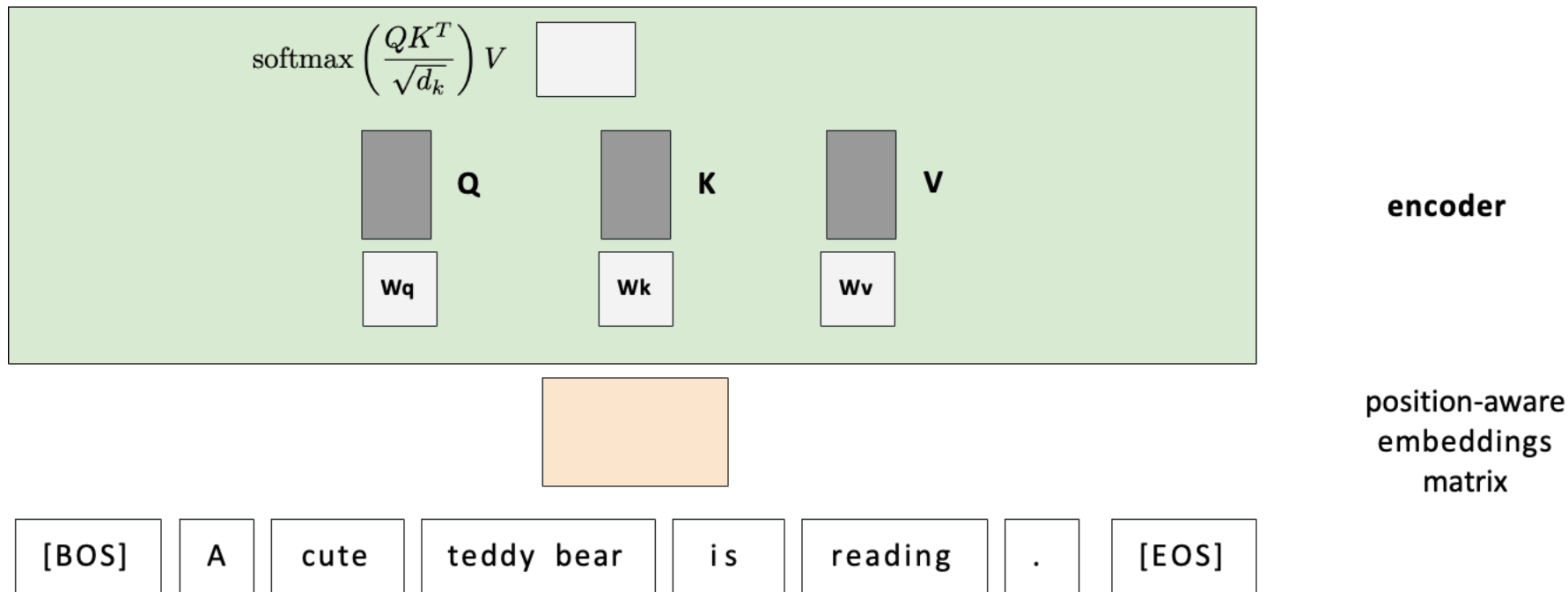
Stitching all the pieces together with an example



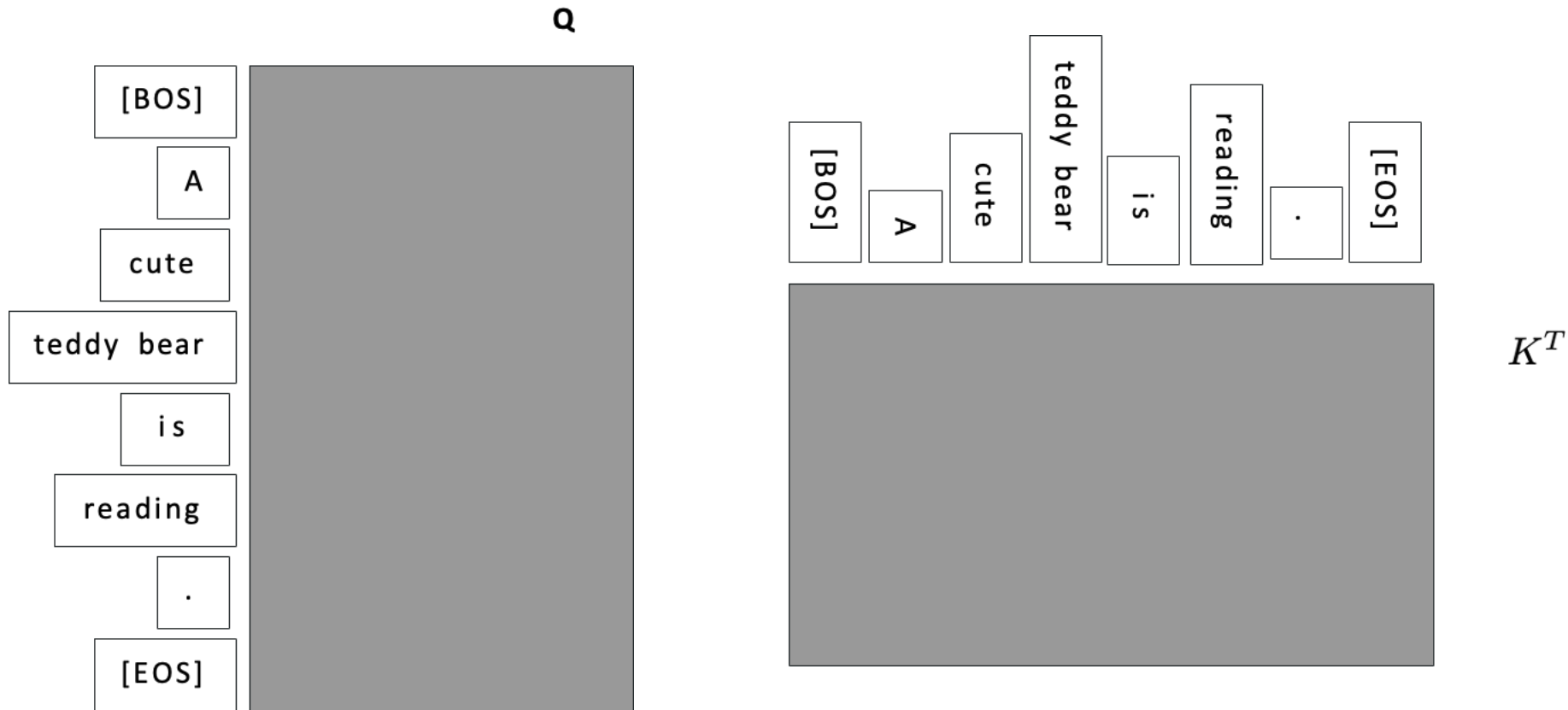
Stitching all the pieces together with an example



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Stitching all the pieces together with an example

$$\langle q_{[\text{BOS}]}, k_{[\text{BOS}]} \rangle \quad \langle q_{[\text{BOS}]}, k_{\text{A}} \rangle \quad \langle q_{[\text{BOS}]}, k_{\text{cute}} \rangle \quad \dots$$

$$\langle q_{\text{A}}, k_{[\text{BOS}]} \rangle \quad \langle q_{\text{A}}, k_{\text{A}} \rangle$$

$$\langle q_{\text{cute}}, k_{[\text{BOS}]} \rangle \quad \begin{matrix} \cdot \\ \cdot \\ \cdot \end{matrix}$$

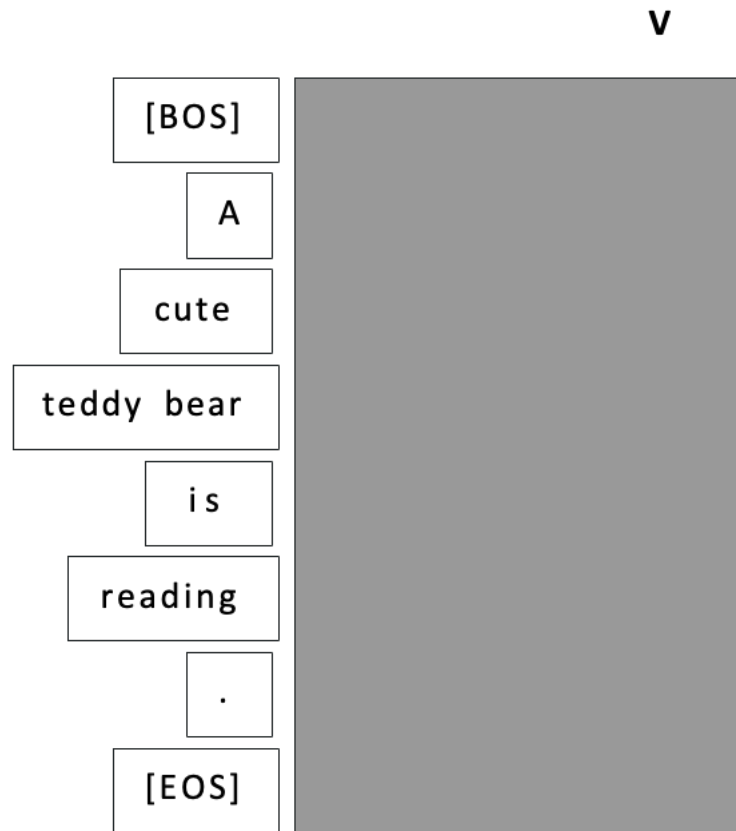
$\vdots$

$$QK^T$$

Stitching all the pieces together with an example

$$QK^T$$

$$\begin{array}{cccc} \langle q_{[\text{BOS}]}, k_{[\text{BOS}]} \rangle & \langle q_{[\text{BOS}]}, k_A \rangle & \langle q_{[\text{BOS}]}, k_{\text{cute}} \rangle & \dots \\ \langle q_A, k_{[\text{BOS}]} \rangle & \langle q_A, k_A \rangle & & \\ \langle q_{\text{cute}}, k_{[\text{BOS}]} \rangle & & \ddots & \\ \vdots & & & \end{array}$$



Stitching all the pieces together with an example

$$\langle q_{[\text{BOS}]}, k_{[\text{BOS}]} \rangle v_{[\text{BOS}]} + \langle q_{[\text{BOS}]}, k_{\text{A}} \rangle v_{\text{A}} + \langle q_{[\text{BOS}]}, k_{\text{cute}} \rangle v_{\text{cute}} + \dots$$

$$\langle q_{\text{A}}, k_{[\text{BOS}]} \rangle v_{[\text{BOS}]} + \langle q_{\text{A}}, k_{\text{A}} \rangle v_{\text{A}} + \langle q_{\text{A}}, k_{\text{cute}} \rangle v_{\text{cute}} + \dots$$

⋮

$$QK^TV$$

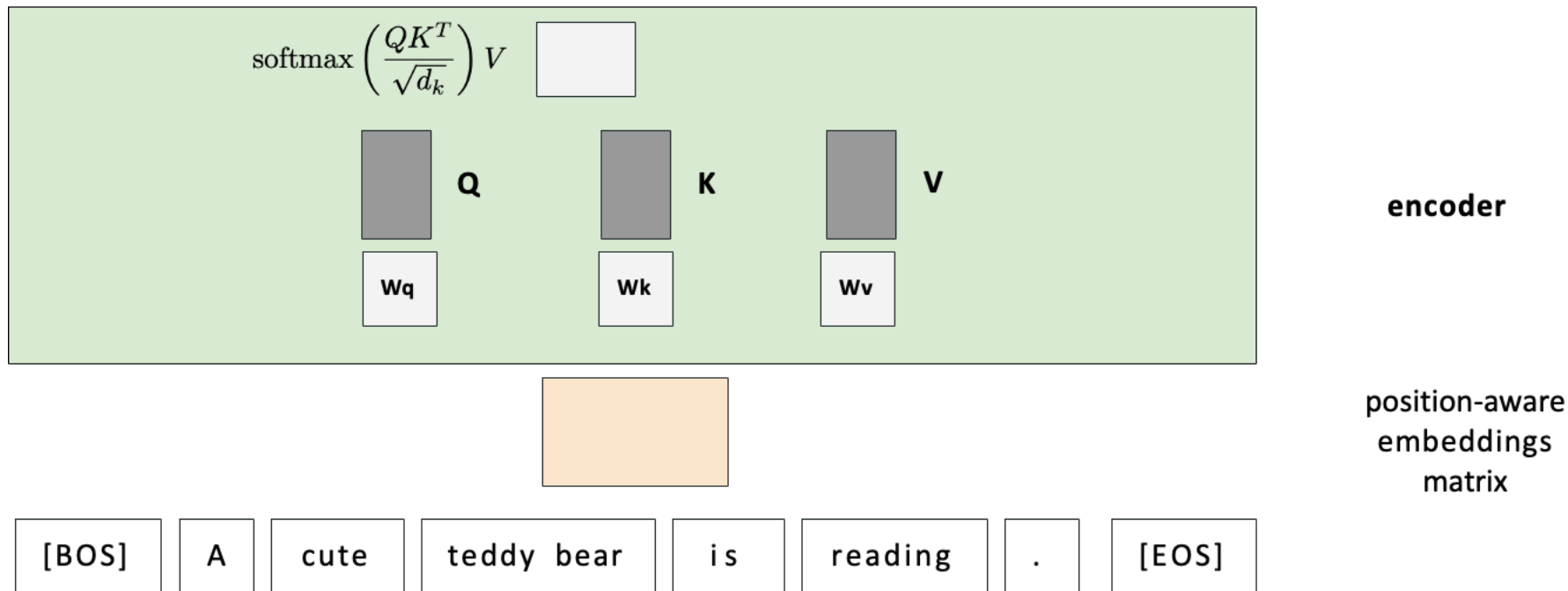
Stitching all the pieces together with an example

$$\text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

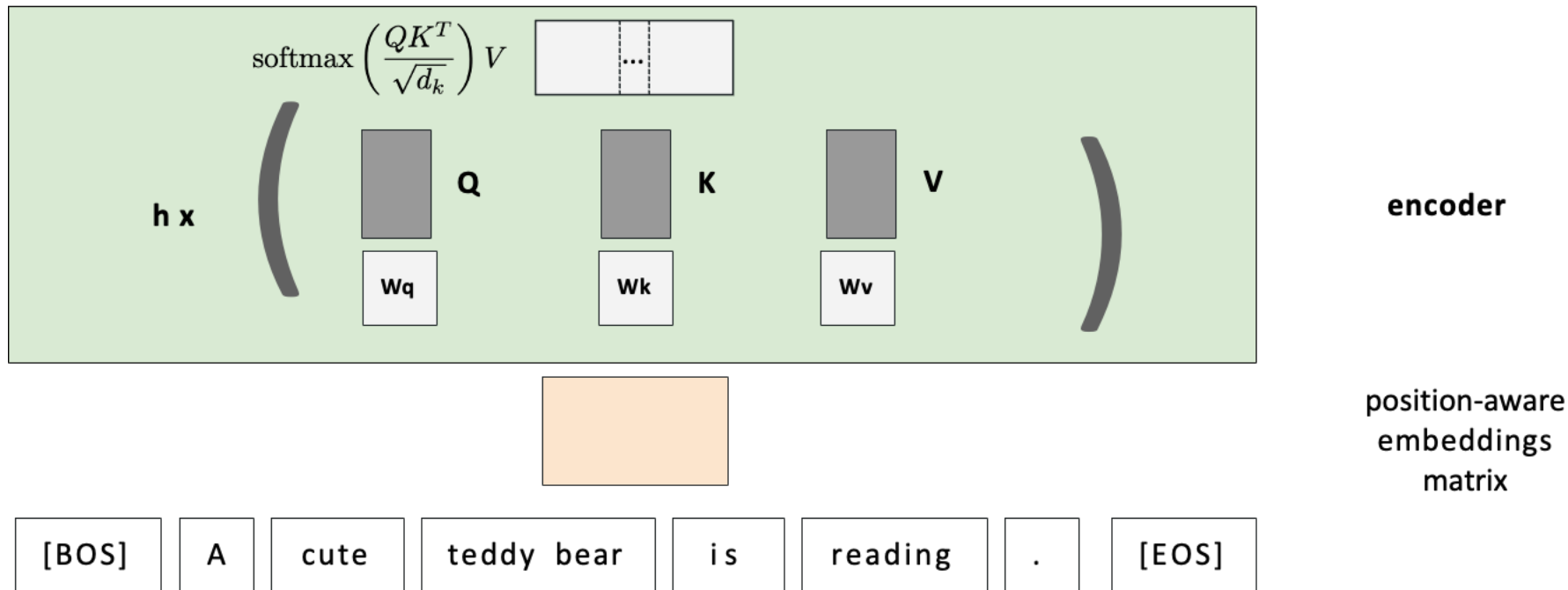
weighted average of values

with weights being a function of $\langle q, k \rangle$

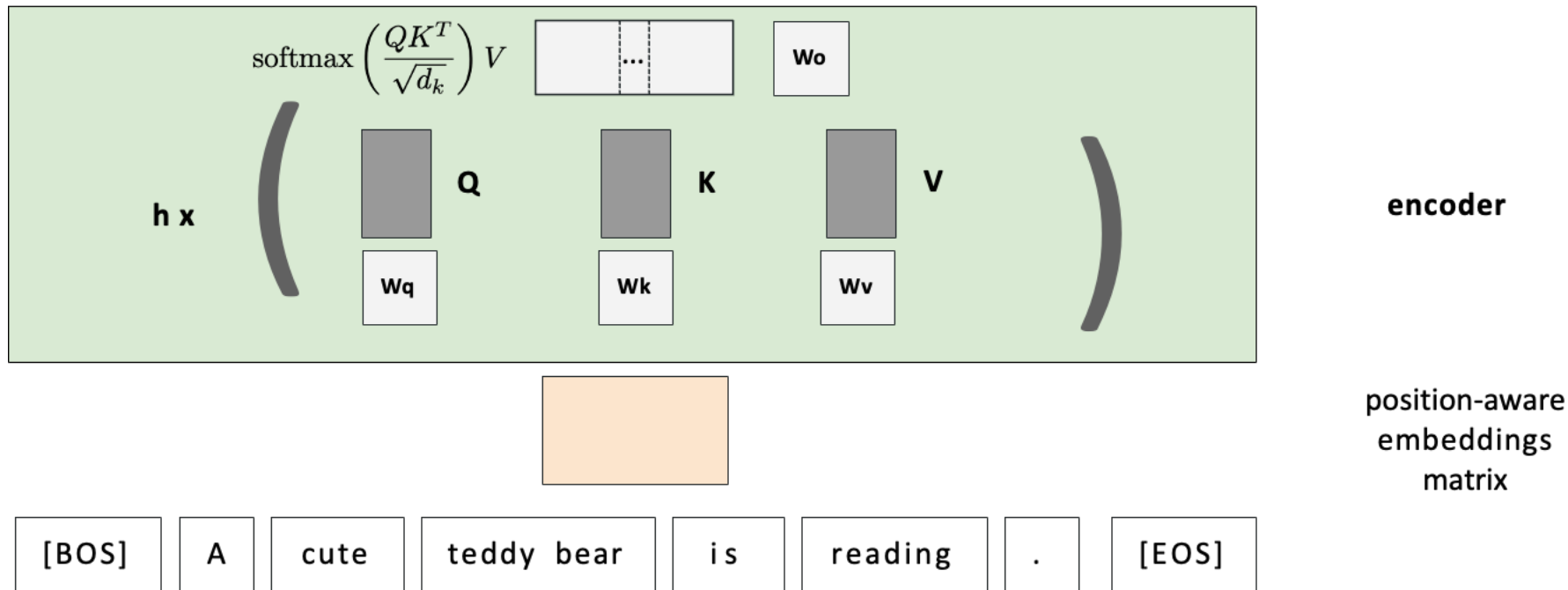
Stitching all the pieces together with an example



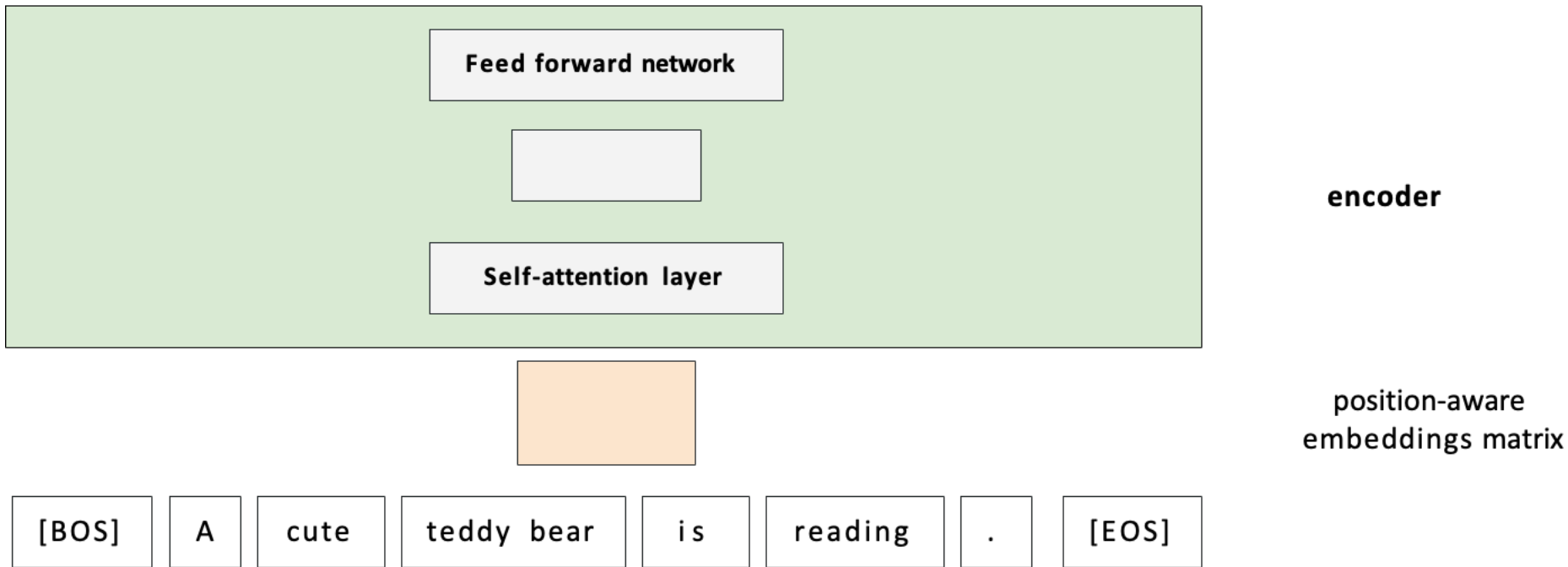
Stitching all the pieces together with an example



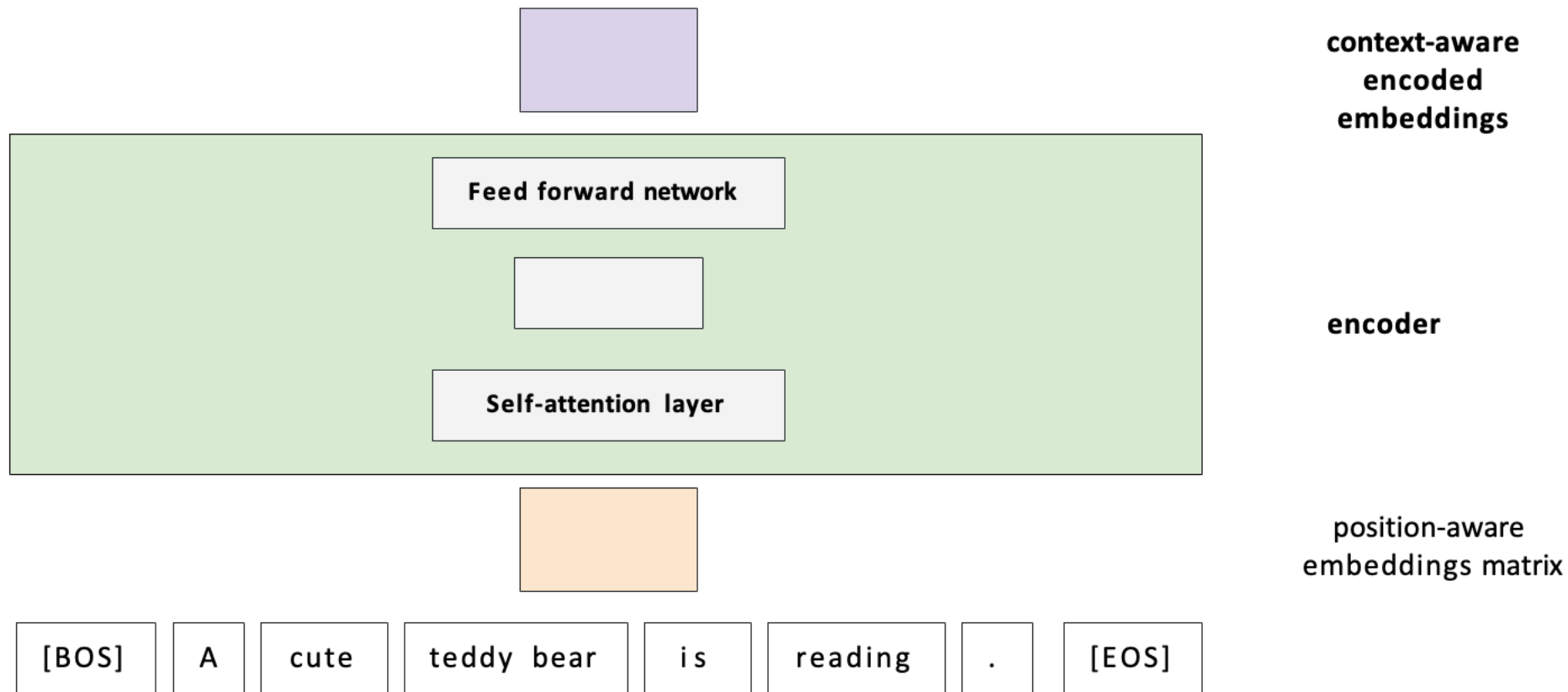
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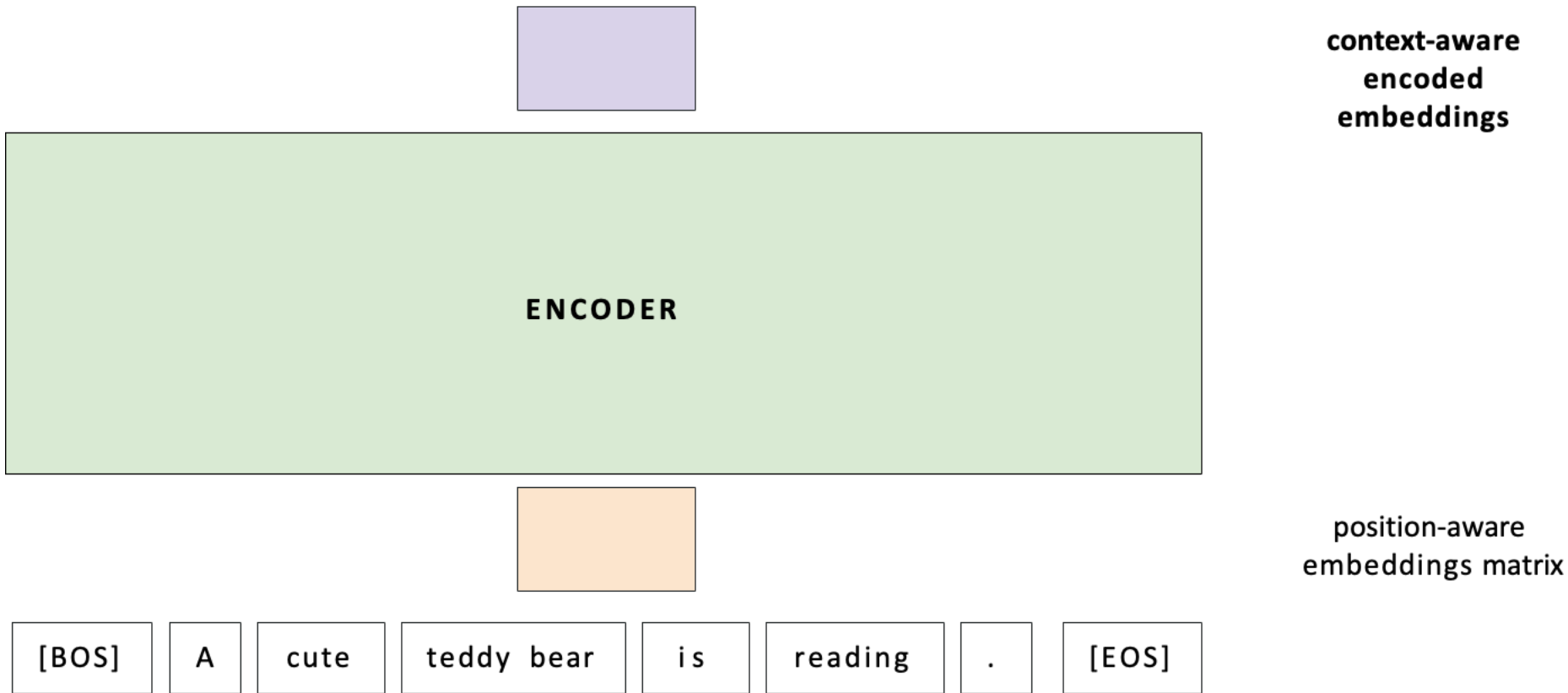
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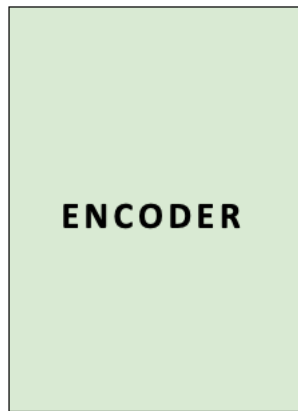
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Stitching all the pieces together with an example



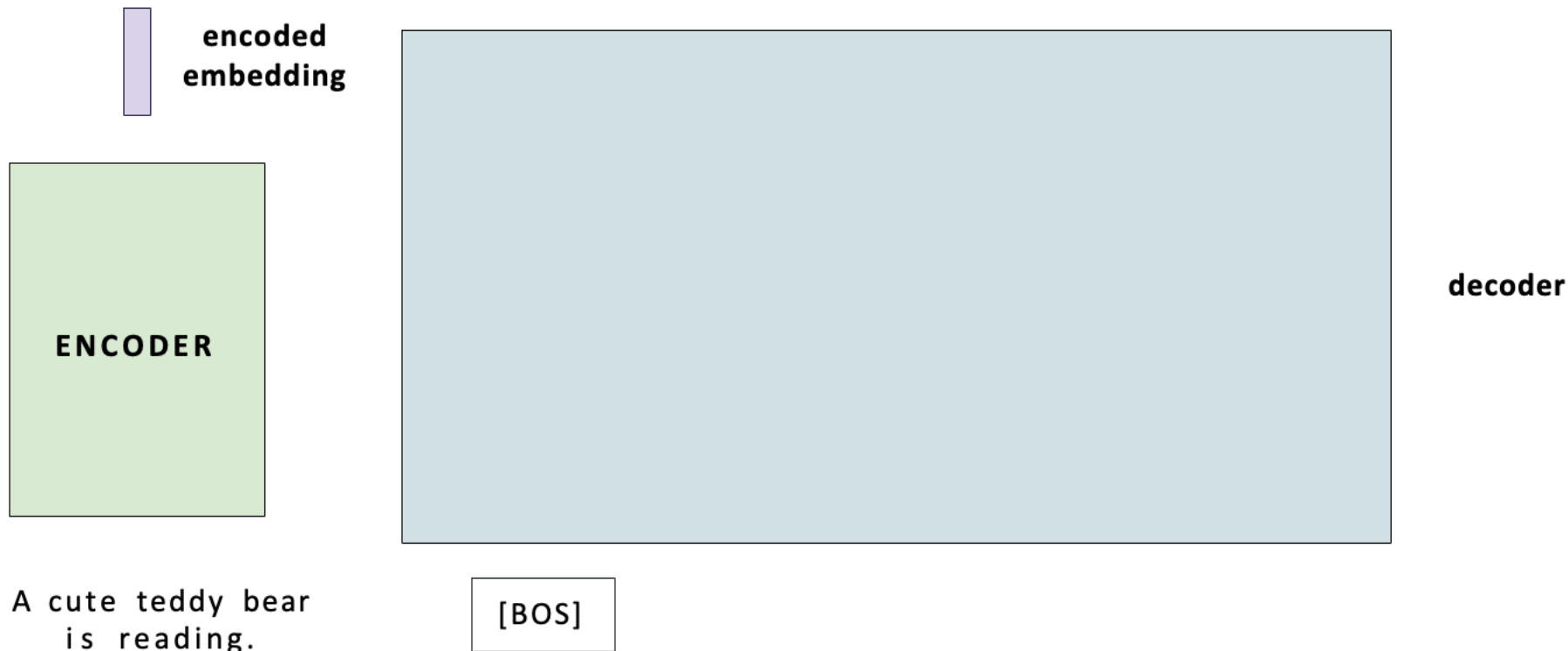
encoded
embedding



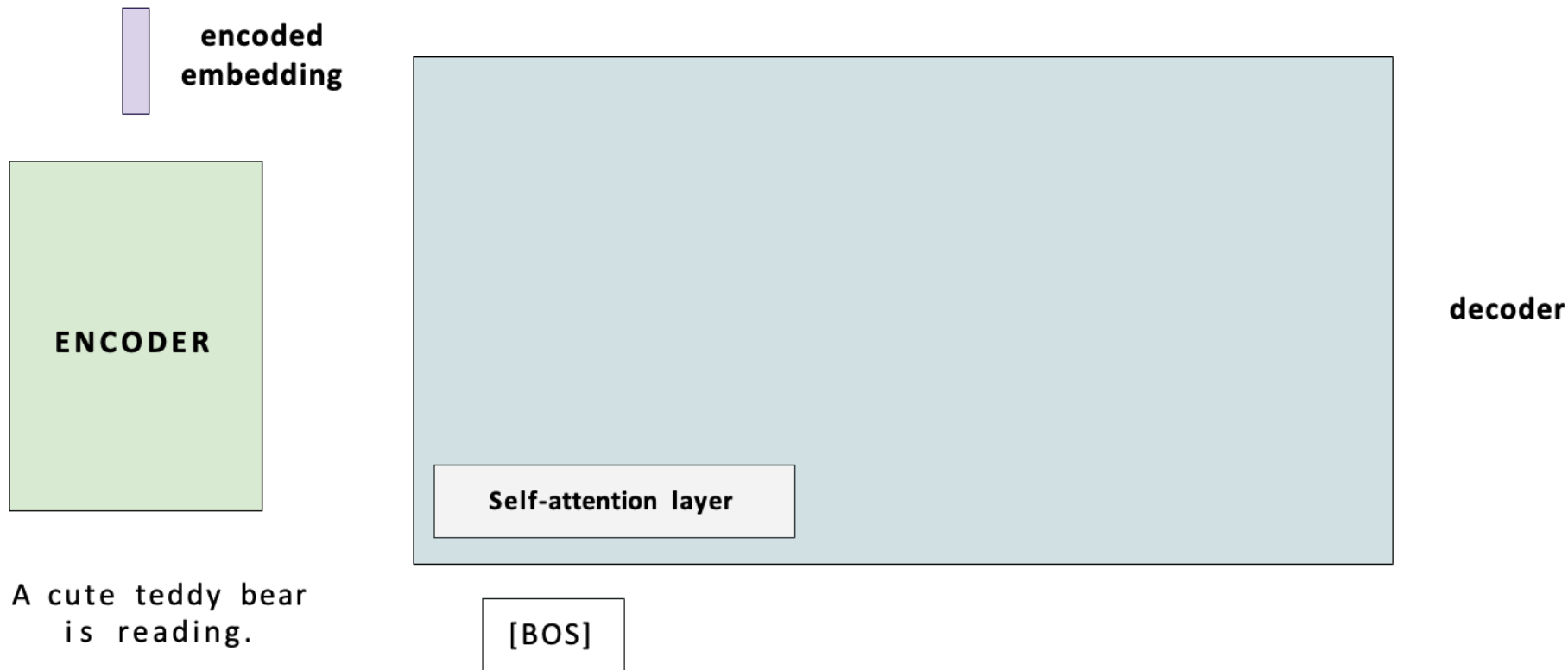
A cute teddy bear
is reading.

[BOS]

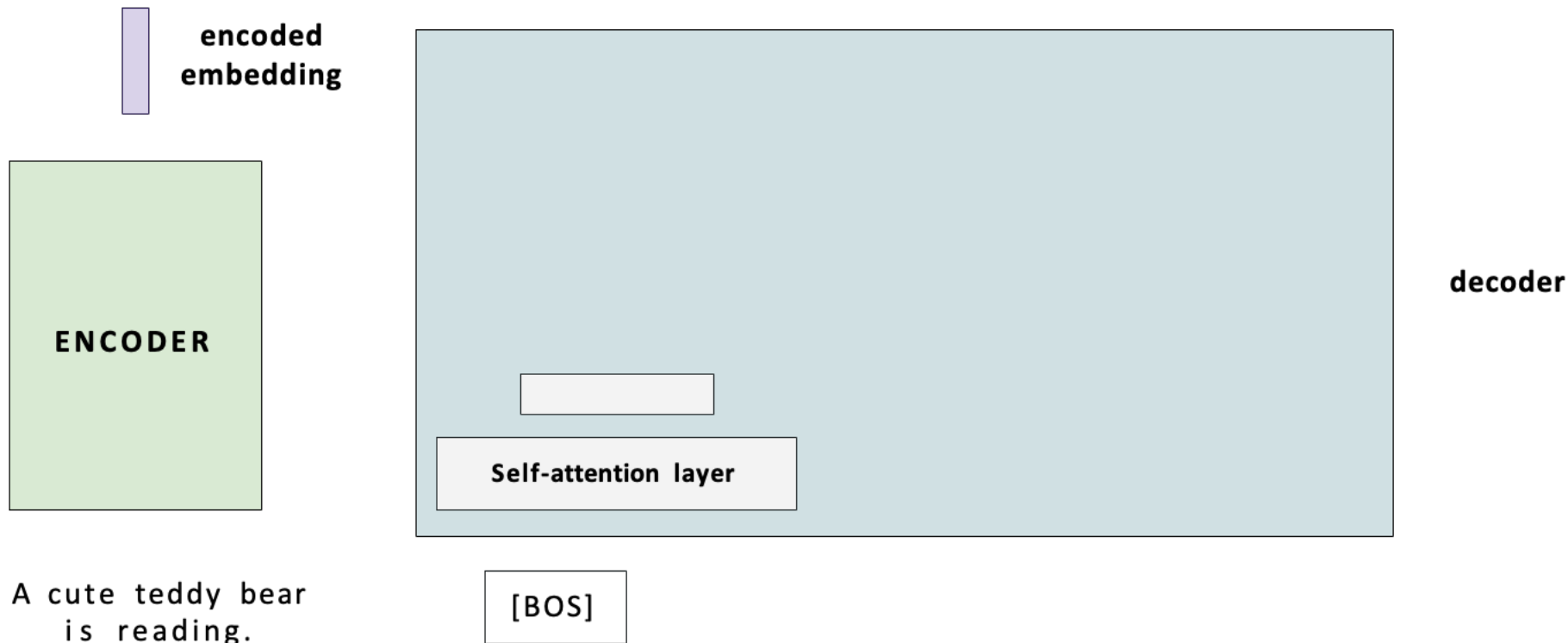
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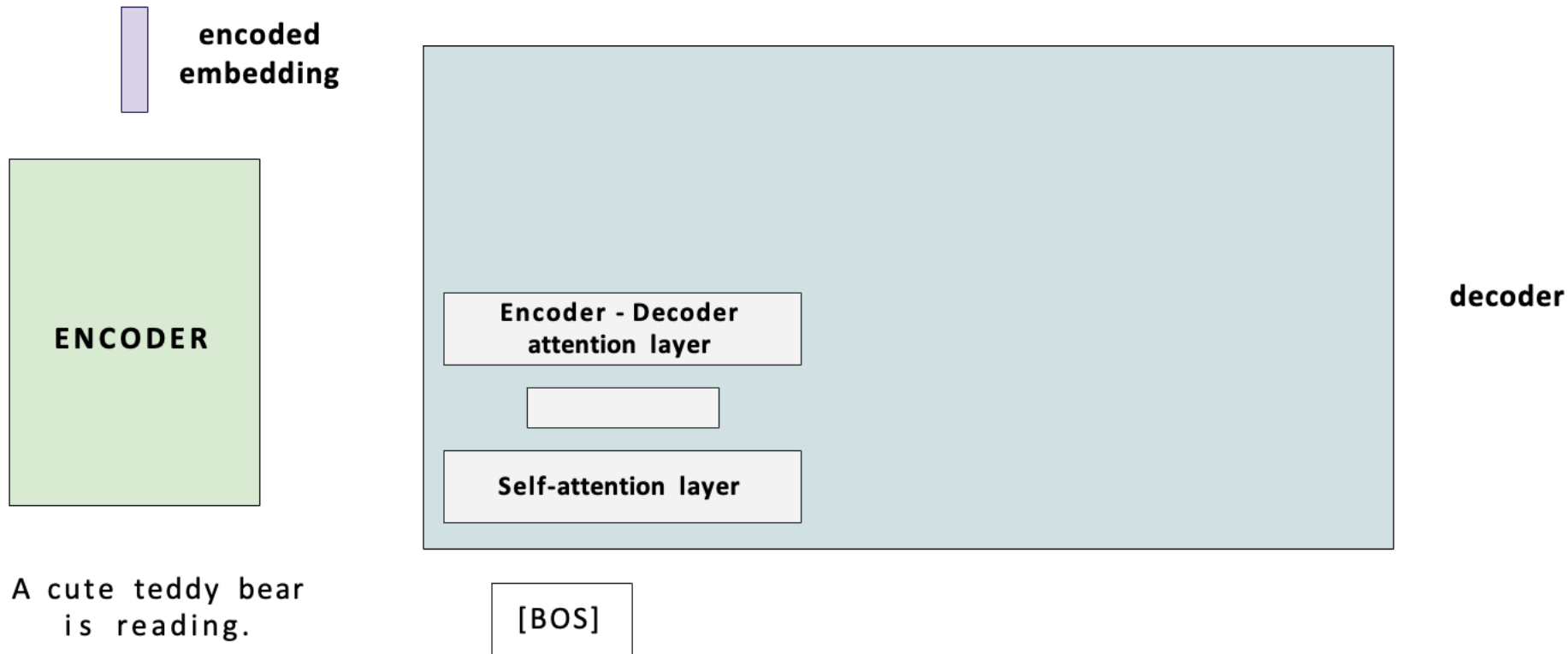
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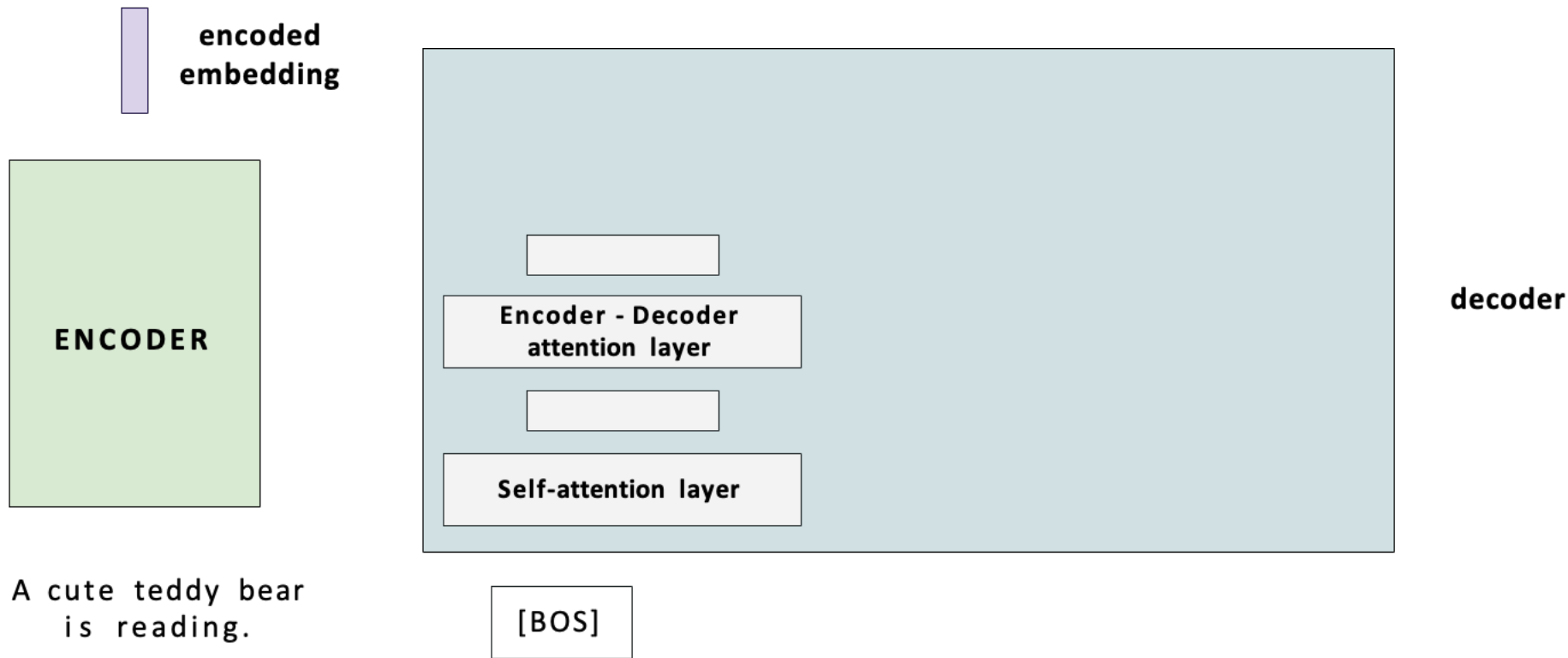
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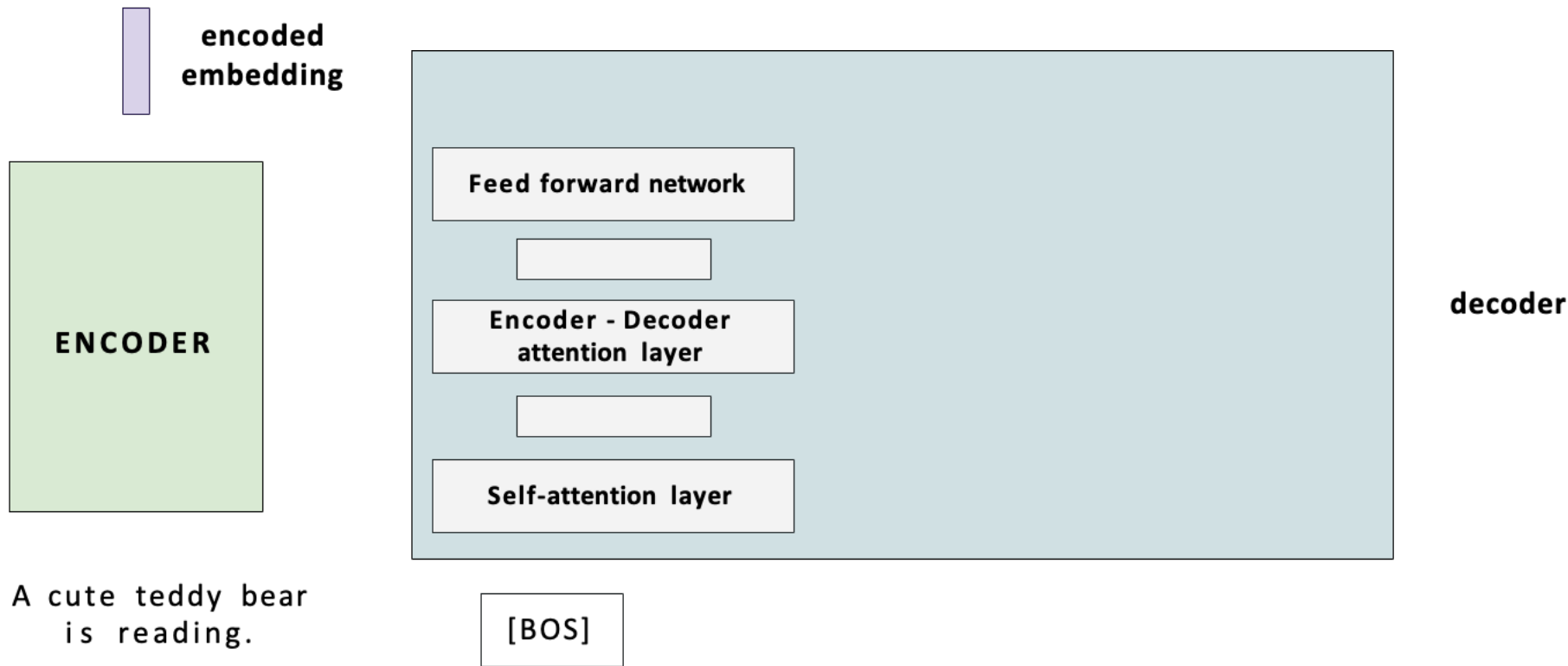
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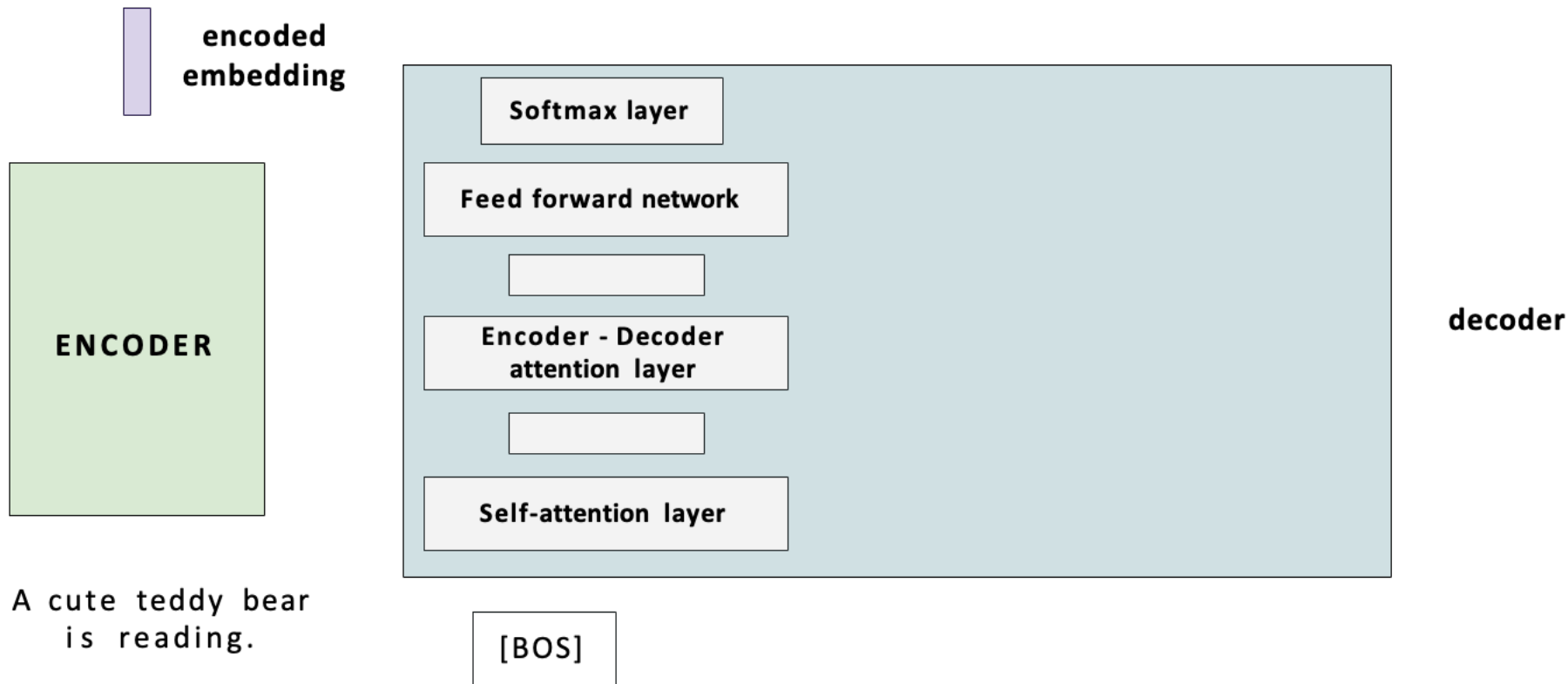
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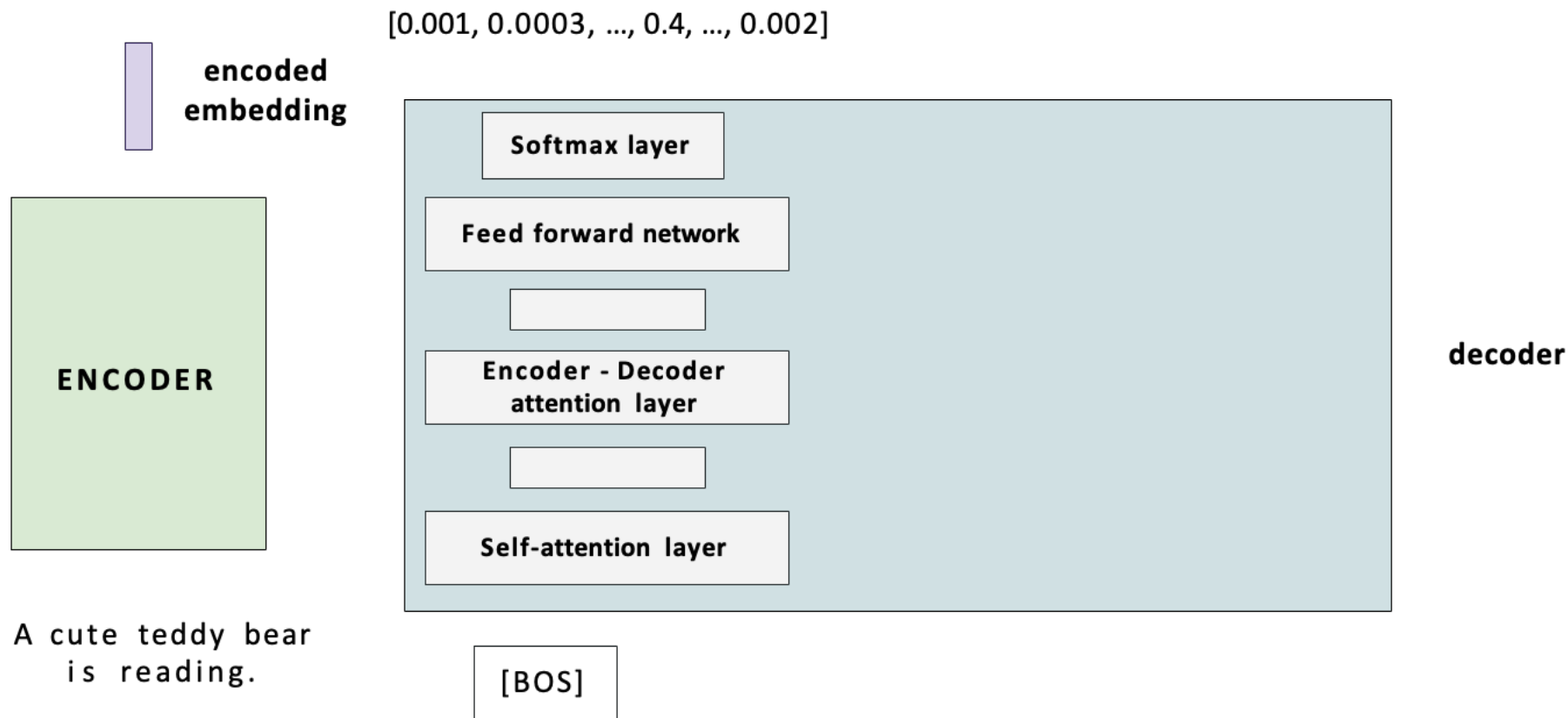
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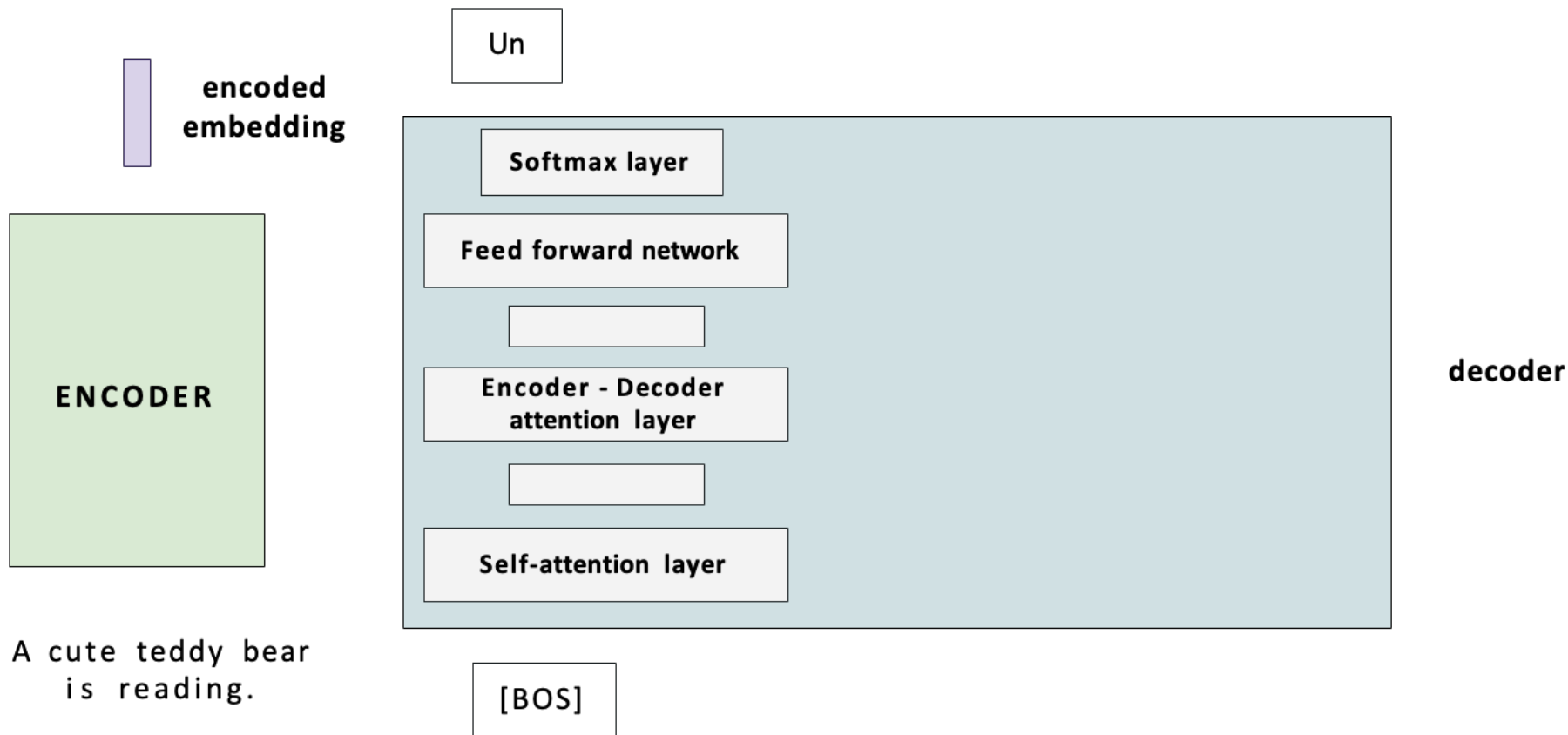
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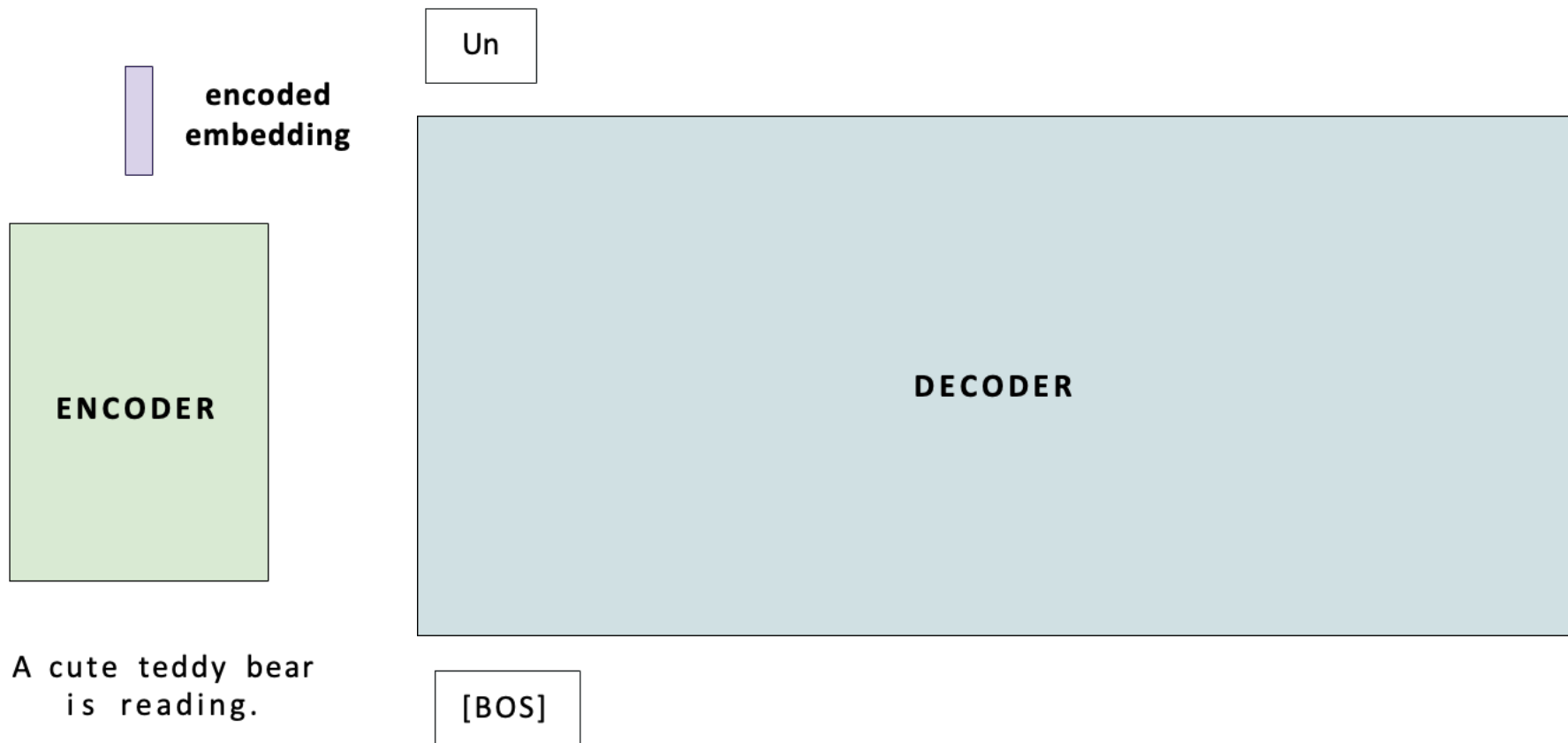
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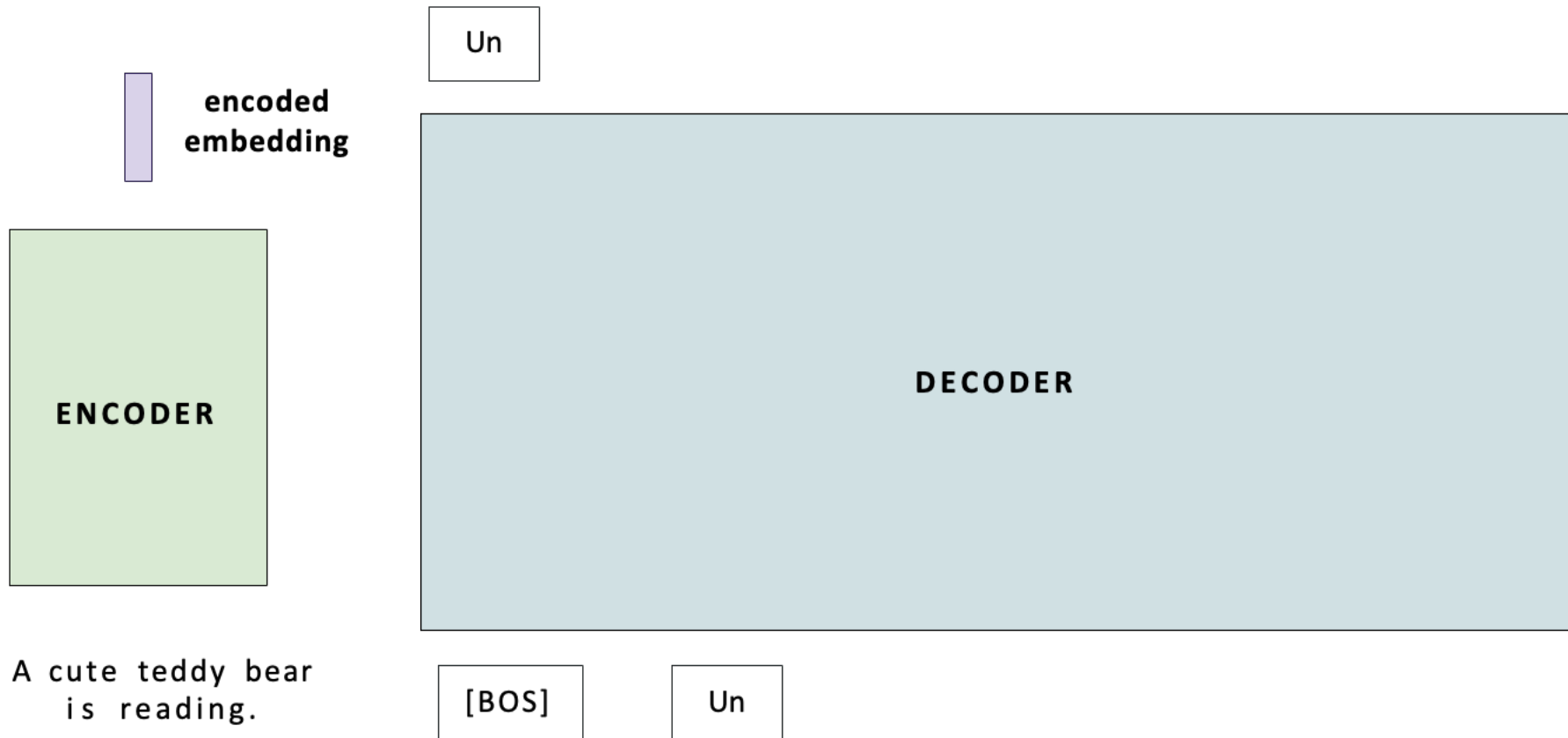
Stitching all the pieces together with an example



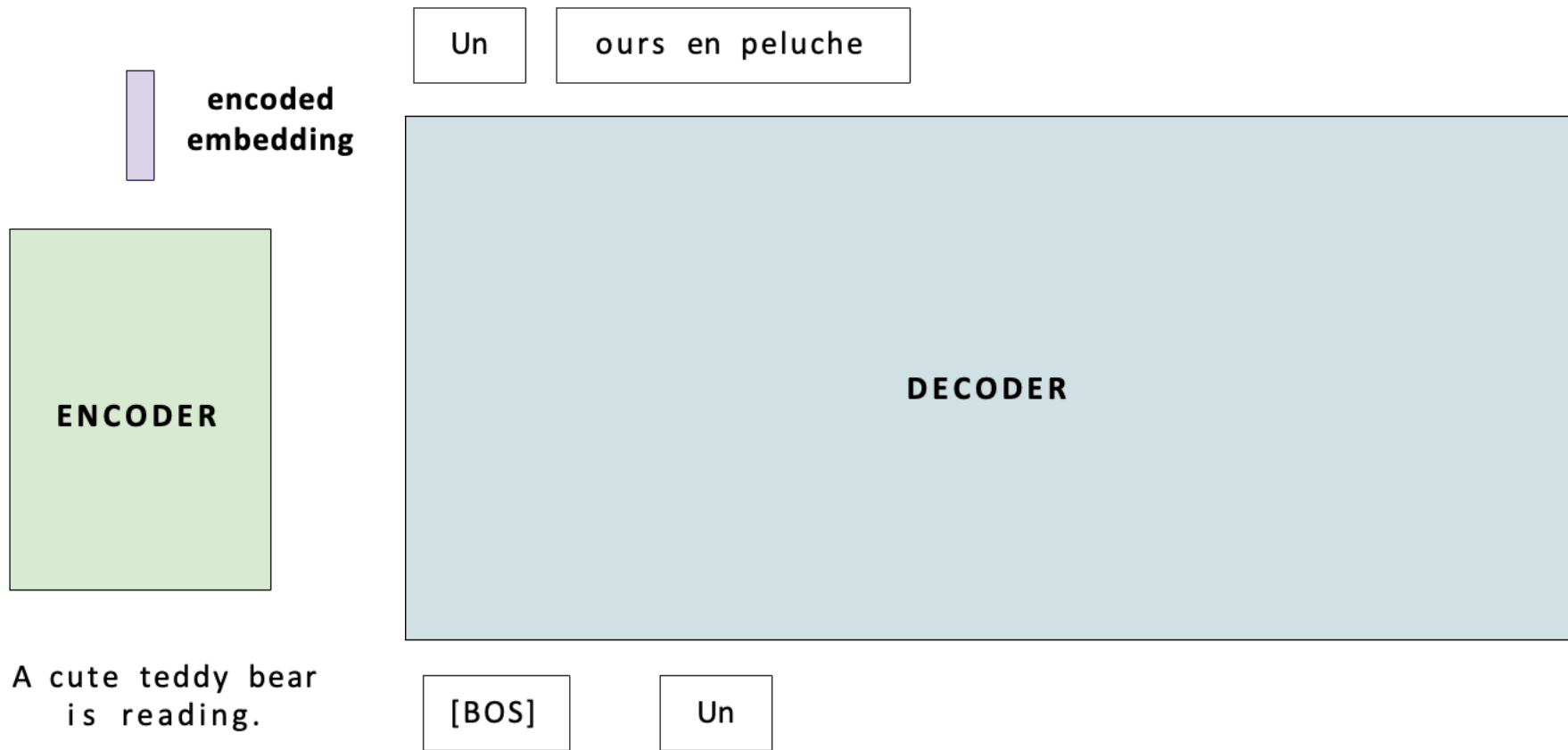
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Stitching all the pieces together with an example



encoded
embedding



Un ours en peluche mignon lit.

ENCODER

DECODER



A cute teddy bear is reading.

Q&A

Any question?